OBJECT DETECTION USING IMAGE PROCESSING

TWO WHEELER HELMET DETECTION USING YOLO V8

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ABSTRACT

Road accidents involving two-wheeler riders are a serious concern, and lack of helmet use often leads to severe injuries. Manual monitoring of helmet compliance is inefficient and prone to human error, creating the need for an automated solution.

This project proposes a real-time helmet detection system using YOLOv8, developed by Ultralytics. The model detects motorcycles, riders, and helmet usage from images and video feeds, even under challenging conditions like poor lighting and complex backgrounds.

The system provides high accuracy, fast detection, and scalability, and can be integrated with intelligent traffic monitoring and automated enforcement to enhance road safety and support smart city development.

INTRODUCTION

Road accidents involving two-wheeler riders are a major global concern, with head injuries being a leading cause of death. Although helmets significantly reduce injury risk, many riders do not follow safety rules.

Advancements in AI and computer vision enable automated traffic monitoring systems. Object detection models like YOLOv8, developed by Ultralytics, provide high-speed and accurate real-time detection.

This project develops a two-wheeler helmet detection system using YOLOv8 to identify motorcycles, riders, and helmet usage from images and videos. The system improves road safety by reducing manual monitoring, increasing enforcement efficiency, and supporting smart city and intelligent transportation systems.

LITERATURE SURVEY

1. A study on helmet detection using CNNs shows better accuracy than traditional image processing methods but faces challenges in real-time use, low lighting, and occlusions.
2. Research on IEEE Xplore highlights a deep learning-based helmet detection system where YOLO offers faster and more efficient performance than region-based methods, though early YOLO versions struggle with small objects and crowded scenes.
3. A paper from SAE International discusses AI-based helmet detection integrated with traffic monitoring and license plate recognition, emphasizing scalable and cost-effective smart city solutions.
4. Recent YOLO advancements, especially YOLOv8 by Ultralytics, improve accuracy, speed, and performance in complex conditions like occlusion and poor lighting.

Image 1 (Top – No Helmet Case)
Detected Object: Rider on a two-wheeler
Class Label: No Helmet ❌
Confidence Score: ~0.87
Bounding Box: Red box around the rider's head
Observation: Rider is clearly not wearing a helmet
System correctly identifies violation
Additional Objects Detected:
Motorcycle
Rider (person)

Image 2 (Bottom – Helmet Case)
Detected Object: Rider on a scooter
Class Label: Helmet ✅
Confidence Score: ~0.88
Bounding Box: Green box around helmet area
Observation:
Rider is wearing a helmet properly
System correctly classifies as safe
Additional Objects Detected:
Scooter
Rider(person)

EXISTING SYSTEM

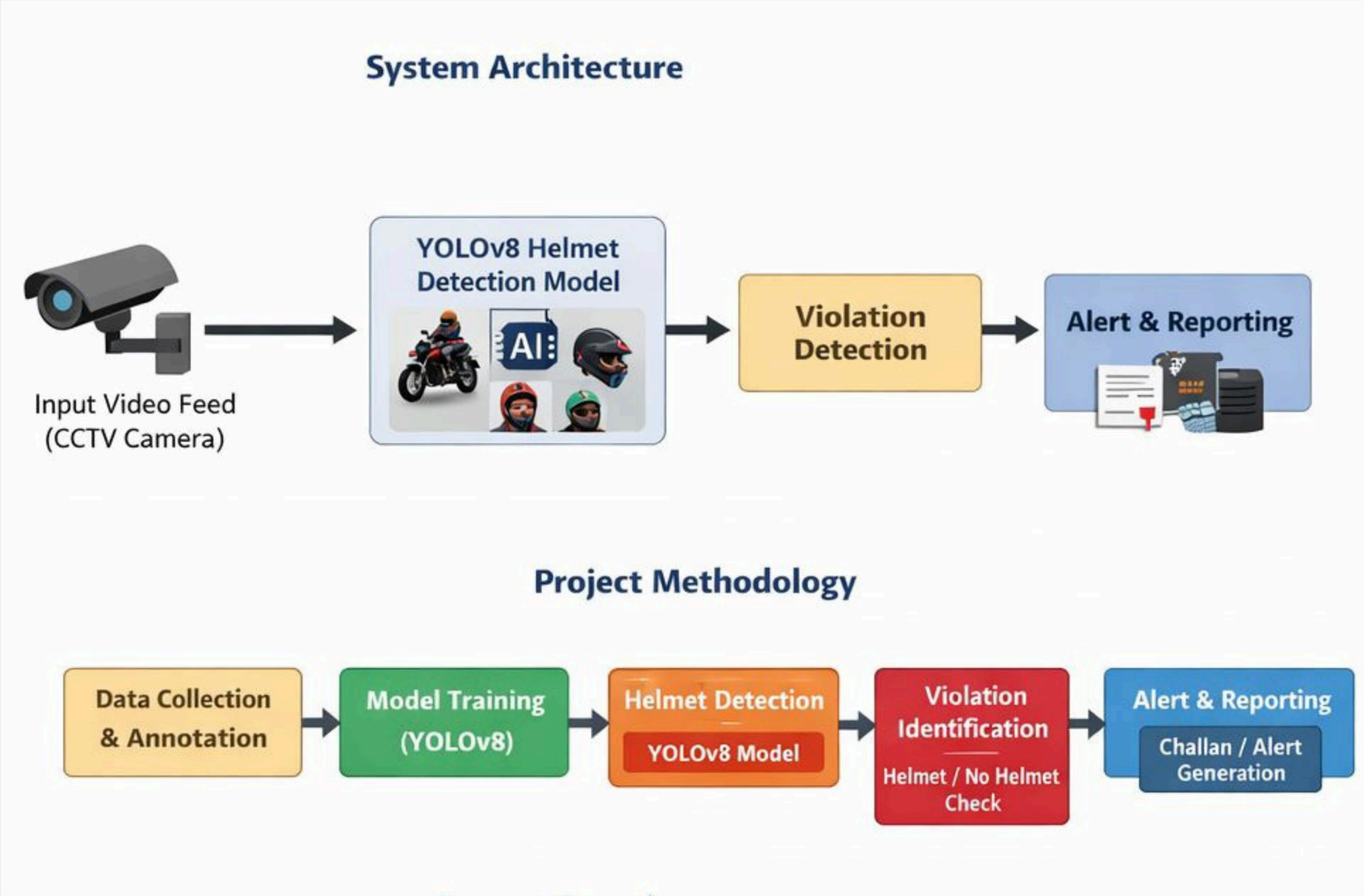
1. Manual Monitoring Systems
   - Traffic police manually observe riders.
   - Time-consuming and inconsistent.
   - Not scalable for large cities.

2. Basic CCTV Surveillance
   - Cameras capture footage but require human review.
   - No real-time violation detection.

3. Traditional Machine Learning Approaches
   - Use handcrafted features for detection.
   - Lower accuracy in complex environments.

4. Early Deep Learning Models (YOLOv5, CNNs)
   - Improved detection but:
   - Struggle with small objects and occlusions
   - High false positives in crowded scenes
   - Limited real-time performance in some cases

BLOCK DIAGRAM



PROPOSED SYSTEM

The proposed system aims to develop a real-time helmet detection model using the YOLOv8 object detection framework. The system is designed to automatically detect whether two-wheeler riders are wearing helmets or not from images and video streams.

A large-scale merged dataset consisting of multiple helmet detection datasets (over 9,000 images) is used to train a deep learning model. The dataset is preprocessed, standardized, and structured in YOLO format for efficient training.

The YOLOv8 medium architecture (YOLOv8m) is used to achieve a balance between accuracy and inference speed. The model is trained using GPU acceleration to optimize performance metrics such as mAP, precision, and recall.

Once trained, the system can:

Detect helmet and no-helmet riders in real-time

Process CCTV footage or live camera streams

Support smart traffic monitoring systems

Assist authorities in automated violation detection

The proposed solution is scalable, efficient, and suitable for integration into smart city surveillance infrastructures.

Violation Detection

- The system detects a rider without a helmet using YOLOv8
- Once confirmed, it marks it as a traffic violation

Alert Generation

- A warning message is generated:
- “No Helmet Detected!”
- Details captured:
- Vehicle number
- Date
- Time

This acts as proof of violation

Vehicle Identification

- The system extracts the vehicle number (KA05MG1234)
- This can be done using:
- Number plate detection (ANPR)

E-Challan Creation

- An automatic challan (fine receipt) is generated:
- Vehicle Number
- Type of Violation
- Fine Amount (₹500)

Payment System

- QR Code is generated for easy payment
- “Pay Now” option allows digital payment

CONCLUSION

The proposed two-wheeler helmet detection system using YOLOv8 effectively applies deep learning to improve road safety by automating helmet monitoring and reducing human error and inefficiency.

YOLOv8, developed by Ultralytics, enables real-time, high-accuracy detection of motorcycles, riders, and helmet usage, even in challenging lighting and complex environments.

The system demonstrates the potential of AI in traffic rule enforcement and can be enhanced with features like license plate recognition and automated penalties, supporting smart city and intelligent transportation systems.

REFERENCE

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Thank you!